

FLOWRRA: Framework Architecture Update

GIBBS-DISTRIBUTED FACTORIZATION & LOCALIZED MAP ESTIMATION

This document formalizes the integration of thermodynamic factorization, Gibbs distributions, and Maximum A Posteriori (MAP) local wave function collapse within the FLOWRRA (Flow Recognition Reconfiguration Agent) architecture. Building upon the previous density estimation and Wave Function Collapse models implemented in Python 3, this framework mathematically deduces the updated cyber-physical middleware required for the ANIMAA substrate, maintaining strict functional separation from the SUTRAA cognitive architecture.

1. The (1 - Gibbs) Obstacle Density Function

To ensure absolute computational frugality, the global state space of a **Holon** (defined strictly as a set of robots) is not modeled around open, viable space. Instead, the Graph Neural Network (GNN) parameterizes the energy landscape solely around physical limits and obstacles.

Let $P_{obs}(X)$ be the Gibbs distribution of failure states (e.g., collisions, joint limits, or signal dead zones in a Starlink satellite constellation loop). The viable density function for the framework is formulated as a repulsive field:

$$P_{flow}(X) \propto 1 - [(1/Z) \times \exp(-E_{obs}(X) / T)]$$

This ensures that when a Holon operates in unconstrained space, compute requirements drop to near zero. The GNN only actively calculates the repulsive energy barrier E_{obs} when a **Node** (defined strictly as one robot or a swarm) approaches a constraint.

2. Local Wave Function Collapse via MAP Estimation

When a Node must act, querying the entire Holon is mathematically intractable. By exploiting the Hammersley-Clifford theorem, the GNN factorizes the global distribution into local clique potentials. A Node collapses into a definitive state using MAP estimation over its local neighborhood $N(i)$:

$$x_i^* = \operatorname{argmax} P(x_i | x_{N(i)}, I)$$

Here, the partition function Z elegantly cancels out. The variable I represents the **Intent Manifold**, injected externally.

3. The Zero-Overhead RL Reward Function (KL Divergence)

The Reinforcement Learning (RL) agent within FLOWRRA does not calculate physics; it learns policy. The GNN dynamically calculates the Kullback-Leibler (KL) Divergence between the Node's collapsed state and the Holon's global Gibbs distribution. The RL agent utilizes this exact metric as its intrinsic reward, reducing computational overhead to $O(1)$ for the policy layer.

$$R_t = -\lambda \times D_{KL} (P_{collapsed} \parallel P_{Gibbs})$$

When divergence spikes (indicating physical drift or signal loss between nodes), the RL agent perceives negative reward and adjusts the Temperature parameter T , inducing localized entropy to repair the link loop.

4. Decentralized Holonic P2P Learning

To avoid single-point-of-failure bottlenecks across distributed setups (e.g., Starlink orbital links), Holons utilize a lightweight Gossip Protocol. When a Holon learns a novel factorization to avoid a new environmental obstacle, it extracts the gradient difference (ΔW) and transmits it to adjacent Holons. Receiving Holons integrate this knowledge using Federated Averaging (FedAvg).

Crucially, this knowledge transfer is clustered by functional similarity. A manipulator Holon (hand) and a locomotion Holon (foot) maintain separate topological learnings, preserving compute efficiency within the ANIMAA substrate.

5. SUTRAA Integration: The Intent Handshake

FLOWRRA operates exclusively as a reactive nervous system. It possesses no inherent cognitive understanding of the environment. Intent is managed by SUTRAA as an external black-box input.

The Intent Manifold

By default, FLOWRRA's energy equation repels Nodes from all physical structures. When SUTRAA establishes an objective (e.g., "grip a cap"), it injects an Intent Manifold ($I_{SUTRAA} = 1$), inverting the local topography from an obstacle (hill) into an attractor (valley):

$$E_{total}(X) = E_{obs}(X) - I_{SUTRAA} \times E_{target}(X)$$

The Energy Cliff Interrupt

When executing a task, such as twisting a cap, FLOWRRA maintains the grip dynamically by balancing the Temperature (T) against the repulsive E_{obs} of the cap. The moment the threads clear and the cap disengages, the physical resistance drops to zero.

This instantaneous structural change in the Gibbs energy landscape triggers an interrupt. FLOWRRA ceases motion and passes a boolean signal to SUTRAA. SUTRAA processes the moment embedding (determining if the object was a cap to be lifted, or a captive bolt to be released) and issues the subsequent Intent Manifold, allowing FLOWRRA to resume its MAP estimation cycle.